SWAMI VIVEKANAND INSTITUTE OF POLYTECHNIC, LATUR (1703)

CLASS	SUBJECT	SUBJECT CODE	DATE	DAY	TIME	MAX. MARKS
EJ3K	CKN	313325				30

Instructions:

1. **Assume suitable data if necessary.**
2. **Figures to the right indicate full marks.**
3. **Use of non programmable electronic pocket calculator is permissible.**
4. **Mobile phone, pager and any other electronic communication devices are not permissible in examination hall.**

Q. 1) Attempt any five out of seven
(2*5=10M)

1. Define the term-

i. Mesh

ii. Node

iii. Branch.

2. Draw-

i. Practical Voltage Source

ii. Ideal Current Source.

3. Draw the Star and Delta Connection and State any One Conversion Formula.

4. State the Need for Source Transformation.

5. Explain with Suitable example to Convert Practical Current Source into Equivalent Voltage Source.

6. State Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL).

7. State and explain Ohm's law

Q. **2) Attempt any four out of seven**
(4*5=20M)

1. Using source conversion, reduce the circuit shown in Fig. into a single voltage source in series with a single resistance.

2. Find the equivalent resistance between terminals A & B for the given circuit.

3. Three resistors 50Ω, 40Ω & 25Ω are connected in parallel. Determine equivalent resistance. If current in 25Ω resistance is 8 A, find current in other two resistances and total power consumed in the circuit
4. Define open circuit and short circuit.
5. Show that using mesh analysis calculate the voltage drop across 10Ω resistor (as shown).
6. Convert the delta network into equivalent star network.